

Auteur: Odia Kibor
Studierichting: Informatica
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$$z^6 = 1$$

$$r = \sqrt{(1)^2 + (0)^2} = 1$$

$$\theta = \text{Bgtan} \left(\frac{0}{1} \right) = 0$$

$$w = \sqrt[n]{r} \text{cis} \left(\frac{\theta + k2\pi}{n} \right)$$

$$k \in 0, 1, 2, 3, 4, 5$$

Wortels

$$w_0 = \text{cis } 0 = 1$$

$$w_1 = \text{cis} \left(\frac{\pi}{3} \right) = \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$w_2 = \text{cis} \left(\frac{2\pi}{3} \right) = \left(\cos \left(\frac{2\pi}{3} \right) + i \sin \left(\frac{2\pi}{3} \right) \right) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$w_3 = \text{cis} (\pi) = (\cos (\pi) + i \sin (\pi)) = -1$$

$$w_4 = \text{cis} \left(\frac{4\pi}{3} \right) = \left(\cos \left(\frac{4\pi}{3} \right) + i \sin \left(\frac{4\pi}{3} \right) \right) = -\frac{1}{2} + \left(-\frac{\sqrt{3}}{2}i \right)$$

$$w_5 = \text{cis} \left(\frac{5\pi}{3} \right) = \left(\cos \left(\frac{5\pi}{3} \right) + i \sin \left(\frac{5\pi}{3} \right) \right) = \frac{1}{2} + \left(-\frac{\sqrt{3}}{2}i \right)$$

References